10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 18 — Salts

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

Quick Check 18.1

1. How the ions present in NaCl arrange themselves to form a crystal lattice?

Sodium chloride is formed when sodium hydroxide neutralises hydrochloric acid, and it consists of Na^+ and Cl^- ions. The oppositely charged ions attract each other strongly and arrange themselves in a regular, repeating three-dimensional pattern called a crystal lattice, which has the shape of a face-centred cube. The bigger chloride ions occupy the corners and the centres of the faces of the cube, while the smaller sodium ions occupy the edges. In this lattice each sodium ion is surrounded by six chloride ions, and each chloride ion is surrounded by six sodium ions.

2. Do you expect melting point of KCl to be higher or lower than NaCl?

The melting point of KCl is lower than that of NaCl. The potassium ion (K^+) is bigger in size than the sodium ion (Na^+). The strength of the electrostatic force of attraction depends upon the size of the ions and the distance between them: as the distance between the ions increases, the attraction becomes weaker. In KCl the ions are farther apart, so a smaller amount of energy is needed to break the lattice, and therefore KCl melts at a lower temperature than NaCl.

Quick Check 18.2

1. Which salts of barium and calcium are soluble in water?

The nitrates of barium and calcium, $Ba(NO_3)_2$ and $Ca(NO_3)_2$, are soluble in water because all metallic nitrates are soluble. Their chlorides, $BaCl_2$ and $CaCl_2$, are also soluble, because chlorides are soluble except those of silver and lead. Calcium hydroxide, $Ca(OH)_2$, is only partially soluble in water. On the other hand, the sulphates and carbonates of both barium and calcium, that is $BaSO_4$, $CaSO_4$, $BaCO_3$ and $CaCO_3$, are insoluble in water.

2. How is barium sulphate prepared in the laboratory?

Barium sulphate is insoluble in water, so it cannot be prepared by the reaction of an acid with a base or a metal; it is prepared by the method of precipitation. A solution of barium chloride is mixed with a solution of sodium sulphate, when a white precipitate of barium sulphate is formed at once. The precipitate is filtered, washed with distilled water to remove the soluble impurities, and dried between the folds of filter paper.

$$BaCl_{2(aq)} + Na_2SO_{4(aq)} \rightarrow BaSO_{4(s)} + 2NaCl(aq)$$

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Exercise (Solved)

A. Multiple Choice Questions

1. Which base is not soluble in water?

(a) KOH (b) $Mg(OH)_2$
(c) NaOH (d) Na_2CO_3

Answer: (b) — Hydroxides are soluble only for sodium, potassium and ammonium (calcium partially); magnesium hydroxide is therefore insoluble in water.

2. The shape of the crystal of NaCl is:

(a) Cubic (b) Hexagonal
(c) Rhombic (d) Trigonal

Answer: (a) — The ions of NaCl are arranged in a face-centred cubic lattice.

3. Which salts are always soluble in water?

(a) Chlorides (b) Sulphates
(c) Nitrates (d) Carbonates

Answer: (c) — All metallic nitrates are soluble in water without any exception.

4. Which salt is formed when marble pieces react with dilute nitric acid?

(a) Sodium nitrate (b) Calcium nitrate
(c) Potassium nitrate (d) Magnesium nitrate

Answer: (b) — Marble is calcium carbonate, so $CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + CO_2 + H_2O$.

5. Water insoluble salt is:

(a) Calcium sulphate (b) Sodium sulphate
(c) Potassium sulphate (d) Magnesium sulphate

Answer: (a) — Sulphates of barium, lead(II) and calcium are insoluble in water; all the other sulphates in the list are soluble.

B. Short Answer Questions

18.1 Which factors are responsible for the strength of electrostatic forces between ions?

The strength of the electrostatic force of attraction between two ions depends upon the magnitude of the charges present on the ions and the distance between them. Bigger charges give a stronger attraction, and smaller ions, being closer to each other, also give a stronger attraction. That is why ionic compounds with highly charged and small ions have very strong lattices and high melting points.

18.2 Is lead (II) chloride soluble in water? Give comments.

Lead(II) chloride is practically insoluble in cold water; it is only slightly soluble. Most chlorides are soluble in water, but the chlorides of lead and silver are the important exceptions to this rule. Lead(II) chloride dissolves more in hot water, but on cooling it separates again as white crystals.

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18.3 Which lead salts are insoluble in water?

Lead(II) chloride ($PbCl_2$), lead(II) sulphate ($PbSO_4$) and lead(II) carbonate ($PbCO_3$) are insoluble in water. Lead(II) nitrate, $Pb(NO_3)_2$, however, is soluble because all nitrates are soluble in water. These insoluble lead salts are therefore prepared by precipitation methods.

18.4 Name two insoluble carbonates.

Calcium carbonate ($CaCO_3$) and magnesium carbonate ($MgCO_3$) are insoluble in water. In general, all carbonates are insoluble in water except the carbonates of sodium, potassium and ammonium, which are soluble.

18.5 What is a crystal lattice?

A crystal lattice is the regular, repeating three-dimensional arrangement in which the positive and negative ions of an ionic compound are packed together by strong electrostatic forces of attraction. In this ordered structure every ion is surrounded by a definite number of ions of opposite charge, and the properties of ionic compounds, such as their high melting points and crystalline nature, are due to this structure.

C. Constructed Response Questions

18.1 Why are melting points of ionic salts generally very high?

In an ionic compound the positive and negative ions are held together in a crystal lattice by very strong electrostatic forces of attraction. A large amount of energy is needed to break these forces and to set the ions free, so that they can move. This energy is supplied by heating, and therefore ionic salts melt only at very high temperatures. The melting point is higher when the charges on the ions are larger and the size of the ions is smaller, because then the attraction is stronger.

18.2 Why do ionic salts exist in the solid state?

The strong electrostatic force of attraction between the oppositely charged ions holds every ion at a fixed position in the crystal lattice. At ordinary temperature the ions do not possess enough energy to break these forces and to move away from their positions. Therefore, under normal conditions of temperature and pressure, ionic compounds exist as hard, brittle, crystalline solids with high melting points.

18.3 How will you prepare calcium sulphate in the laboratory?

Calcium sulphate is insoluble in water, so it is prepared by the method of precipitation, that is, by mixing the solutions of two soluble salts which react to give an insoluble product. A solution of calcium chloride is mixed with a solution of sodium sulphate, and a white precipitate of calcium sulphate is formed. The precipitate is filtered, washed with distilled water and dried between folds of filter paper.

$$CaCl_{2(aq)} + Na_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + 2NaCl(aq)$$

18.4 Compare the melting points of NaCl and $MgCl_2$.

NaCl	$MgCl_2$
Sodium ion has a small charge of +1.	Magnesium ion has a bigger charge of +2.
The sodium ion is comparatively larger in size.	The magnesium ion is smaller in size.
The bond is mainly ionic in nature.	Greater polarisation gives the bond some covalent character.
Measured melting point is about 801 °C.	Measured melting point (anhydrous) is about 714 °C.
It is a typical ionic compound with a simple lattice of Na^+ and Cl^- ions.	Its higher charge shortens the distance between ions, but polarisation reduces the ionic character.

18.5 How would you prepare pure crystals of copper sulphate in the laboratory?

Copper sulphate is a soluble salt, so it is prepared by the reaction of an acid with an excess of an insoluble base. A measured volume of dilute sulphuric acid is warmed in a beaker, and black copper(II) oxide powder is added with constant stirring until some of the powder remains undissolved, which shows that all the acid has reacted. The mixture is filtered to remove the excess copper oxide, and the blue filtrate is heated gently to concentrate it until a thin film of crystals appears on the surface. The solution is then cooled slowly, when blue crystals of hydrated copper sulphate separate out. These crystals are filtered, washed with a little cold distilled water and dried between folds of filter paper.

$$CuO(s) + H_2SO_{4(aq)} \rightarrow CuSO_{4(aq)} + H_2O(l)$$

D. Descriptive Questions

18.1 Explain the formation of sodium chloride crystal.

When sodium hydroxide neutralises hydrochloric acid, sodium chloride and water are formed. The compound consists of Na^+ and Cl^- ions, which are held together by the electrostatic force of attraction.

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

Because opposite charges attract each other strongly, the ions arrange themselves in a regular and repeating three-dimensional pattern called a crystal lattice. The lattice of sodium chloride has the shape of a face-centred cube: the bigger chloride ions are present at the corners and at the centre of each face of the cube, while the smaller sodium ions occupy the edges. In this arrangement each sodium ion is surrounded by six chloride ions, and each chloride ion is surrounded by six sodium ions. It is this ordered packing of ions, held by strong electrostatic forces, which gives sodium chloride its crystalline shape, its hardness and its high melting point.

18.2 Explain the methods of preparation of two soluble salts.

(i) Titration method – reaction of a soluble acid with a soluble base: this method is used for salts whose base and acid are both soluble in water. For example, potassium nitrate is prepared by reacting potassium hydroxide solution with dilute nitric acid, using phenolphthalein as indicator, until the pink colour just disappears. The neutral solution is then evaporated to the point of crystallisation, cooled slowly so that crystals of potassium nitrate grow, and the crystals are filtered and dried.

$$KOH(aq) + HNO_{3(aq)} \rightarrow KNO_{3(aq)} + H_2O(l)$$

(ii) Excess insoluble base or metal method: in this method the insoluble base, carbonate or metal is added in excess to the warm acid, so that the whole of the acid reacts. The excess solid is then removed by filtration, and the filtrate is evaporated to a saturated solution and cooled to obtain crystals. For example, zinc sulphate is prepared by adding excess zinc granules to warm dilute sulphuric acid, and copper sulphate by adding excess copper oxide to warm dilute sulphuric acid.

$$Zn(s) + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}$$

18.3 How are the following salts prepared? K_2SO_4, $CuCl_2$, $CaCl_2$

Potassium sulphate is a soluble salt of a soluble base, so it is prepared by the titration method. Potassium hydroxide solution is titrated against dilute sulphuric acid until the solution becomes neutral, and then the solution is evaporated and cooled to get crystals of the salt.

$$2KOH(aq) + H_2SO_{4(aq)} \rightarrow K_2SO_{4(aq)} + 2H_2O(l)$$

Copper(II) chloride is prepared by the reaction of warm dilute hydrochloric acid with an excess of insoluble copper(II) hydroxide. The mixture is filtered to remove the unreacted base, and the filtrate is evaporated and cooled to obtain crystals.

$$Cu(OH)_{2(s)} + 2HCl(aq) \rightarrow CuCl_{2(aq)} + 2H_2O(l)$$

Calcium chloride is prepared by adding excess marble chips (calcium carbonate) to dilute hydrochloric acid. When the reaction stops, the leftover marble is filtered off and the filtrate is evaporated and cooled to get crystals of calcium chloride.

$$CaCO_{3(s)} + 2HCl(aq) \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O(l)$$

18.4 Describe which salts are soluble in water and which are insoluble.

The solubility of a salt in water can be predicted from the general solubility rules, which are summarised in the following table.

Type of salt	Soluble	Insoluble
Sodium, potassium and ammonium salts; all nitrates	All are soluble	None

Type of salt	Soluble	Insoluble
Chlorides, bromides and iodides	Most of them are soluble	Those of silver and lead(II)
Sulphates	Most of them are soluble	Those of barium, lead(II) and calcium
Carbonates	Those of sodium, potassium and ammonium	All the others are insoluble
Hydroxides	Those of sodium, potassium and ammonium (calcium partially)	All the others are insoluble

By using these rules it is possible to decide whether a salt can be prepared by the reaction of an acid with a base, metal or carbonate, or whether it must be prepared by precipitation from two soluble salts.

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